

TECHNICAL PROJECT SUMMARY: ML INTERNSHIP

Institution: Win Research Centre (WRC)

Project: Real-Time Image Classification for the Fit Choice World (FCW) Application

Role: Machine Learning Intern

Duration: December 2024 – March 2025

PROJECT OBJECTIVE

The objective was to architect a high-accuracy, lightweight image classification system to automate meal and exercise logging for the FCW fitness ecosystem. The system categorizes user-uploaded images into three specific classes: Food, Fitness, and Random/Miscellaneous (unrelated media filtering).

TECHNICAL ARCHITECTURE & DEPLOYMENT

- **Model Core:** Leveraged MobileNetV2 (Transfer Learning) via TensorFlow/Keras for its efficiency on resource-constrained devices.
- **Optimization Layer:** Converted the trained model to TensorFlow Lite (TFLite), achieving a compact 3.7 MB footprint and real-time on-device inference.
- **Web Integration:** Developed a responsive Flask-based web interface to manage image buffers and provide instantaneous category predictions.

ENGINEERING CHALLENGES & SOLUTIONS

- **Overfitting & Bias Mitigation:** Faced initial overfitting in the 'Random' and 'Food' categories. Resolved this by implementing Data Augmentation techniques (rotation, scaling, and flips) and expanding the dataset during Week 5 to improve generalization.
- **Production Deployment:** Debugged Flask file-upload routes to handle diverse image formats and optimized the TFLite inference pipeline to maintain a ~30ms latency threshold.
- **Hyperparameter Tuning:** Refined model weights and parameters during Week 6 to fix category-specific misclassifications, resulting in a finalized accuracy of 97%.

KEY SKILLS & OUTCOMES

- **Evaluation Score:** 100/100
- **Core Competencies:** Proficiency in TFLite Optimization, Transfer Learning, Flask Backend Development, and Design-driven AI implementation.
- **Conclusion:** Successfully delivered a production-ready AI solution that balances high precision with extreme computational efficiency, laying the foundation for intelligent fitness applications.

RESULTS

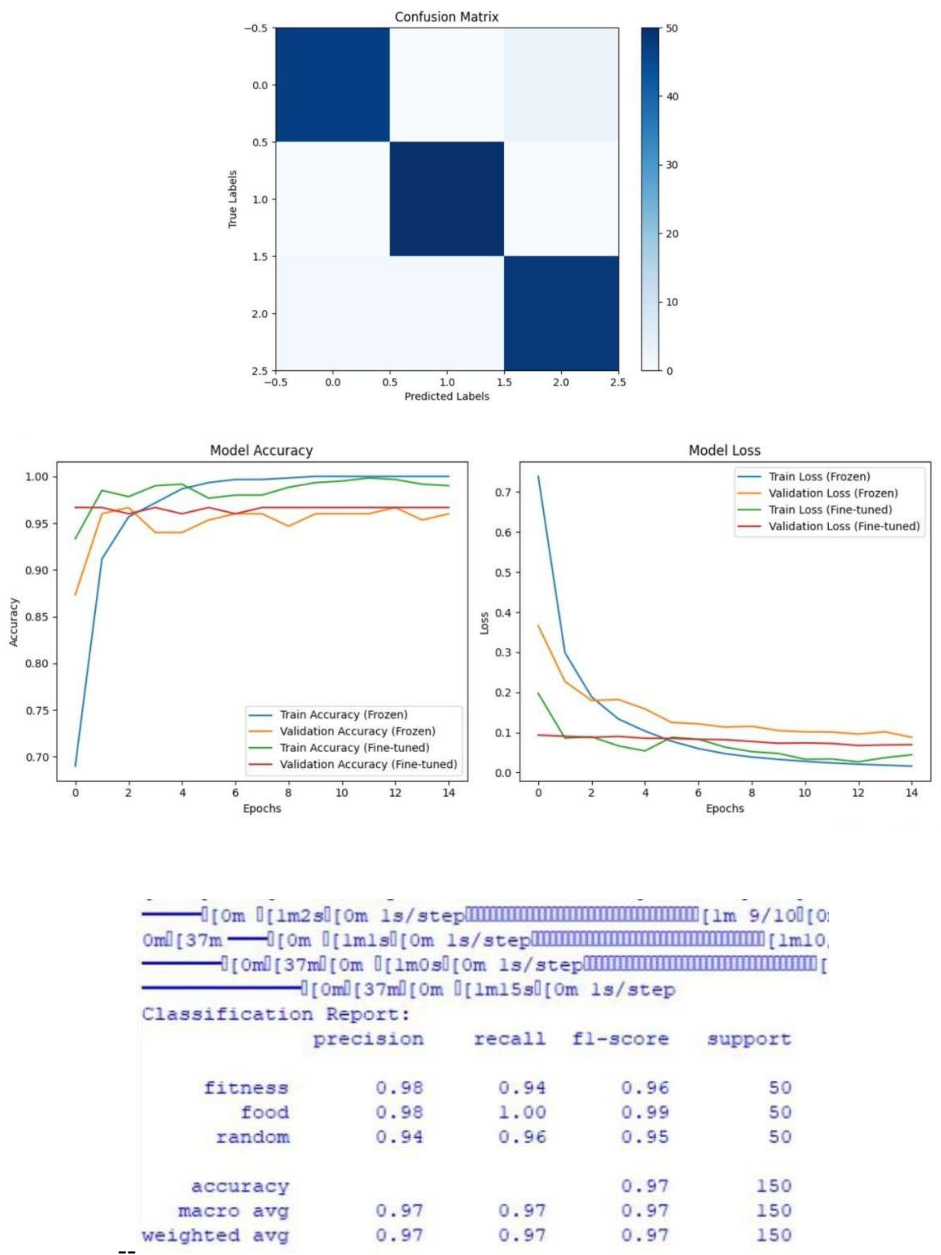


IMAGE CLASSIFICATION

FLASK WEB INTERFACE

